

EVs, Robotics, Semiconductors, Tariffs, and Industrial Policy

A student workbook for turning a market story into a value-chain thesis.

Audience	Beginner-to-intermediate market students who are learning how to research themes without chasing headlines.
Core skill	Map the full value chain: demand, margins, policy, supply chain, competitive position, and company evidence.
Current as of	April 21, 2026

Teaching principle

Do not teach this as 'EVs are dead' or 'robotics is guaranteed.' Teach it as a research process: when a theme matures, the better question becomes where the next bottleneck, margin pool, or policy advantage lives.

Learning objectives

- Separate a headline theme from the businesses that monetize it.
- Explain why global EV sales can still rise while some EV companies face margin pressure.
- Map robotics as a stack: chips, sensors, actuators, batteries, software, factories, and customers.
- Understand why tariffs and domestic industrial policy can change competitive outcomes.
- Build a thesis worksheet using primary evidence instead of social-media confidence.

1. Conversation anchor

In the April 21, 2026 session, the market discussion moved from electric vehicles to robotics. The key teaching point was that students should not only ask whether Tesla sells cars, but how a company may convert manufacturing, engineering talent, AI infrastructure, and supplier relationships into the next product cycle. The same section also connected China competition, tariffs, GPUs, ASML, TSMC, Nvidia, and Tesla manufacturing conversion.

Use this anchor to teach students that a stock is not just a ticker. It is a node inside a larger system.

Instructor framing

The student should leave this tutorial able to say: 'I am not buying an EV headline. I am mapping the full industrial stack and deciding where the evidence is strongest.'

2. The core model: from obvious ticker to value chain

The obvious trade is often the front-facing product: the car, the robot, the phone, or the app. The stronger research habit is to map the layers underneath that product. A student who only studies the car company may miss the battery supplier, chip supplier, foundry, lithography provider, factory automation company, or policy moat.

Layer	Question students must ask	Example evidence
End product	What does the customer actually buy?	EV units, robot deployments, fleet orders, subscription revenue.
Critical inputs	What parts constrain production or margins?	Batteries, motors, cameras, sensors, actuators, memory, GPUs/ASICs.
Compute layer	What silicon and software make the product smarter?	Nvidia platforms, custom AI chips, foundry capacity, edge inference.
Manufacturing	Can the company scale cheaply and reliably?	Capex, factory footprint, automation, labor, supplier concentration.
Policy layer	Does government make the product cheaper, harder, protected, or restricted?	Tariffs, tax credits, safety rules, export controls, domestic manufacturing grants.
Profit pool	Where does durable margin live?	Software, services, replacement parts, data, platform lock-in, recurring subscriptions.

3. EVs are not dead - the lesson is maturity and margin pressure

A clean way to teach the EV debate is to avoid extremes. The International Energy Agency reported that electric car sales increased by more than 20% year over year in 2025 to 21 million units, or about one in four cars sold globally [S1]. That data supports the idea that EV adoption is still growing globally. It does not prove that every EV manufacturer will have attractive margins.

Students should learn the difference between demand growth and investment return. A growing market can still be brutal if competition forces lower prices, incentives disappear, capacity is overbuilt, or the profit pool shifts to batteries, software, chips, charging, or financing.

Signal	Bullish interpretation	Bearish interpretation	Student action
Unit sales up	Product category is expanding.	Growth may be concentrated in lower-margin regions or subsidized segments.	Break sales down by region and price tier.

Signal	Bullish interpretation	Bearish interpretation	Student action
Price cuts	Company is expanding adoption.	Competition is pressuring margins.	Compare gross margin, ASPs, and incentive levels.
Tariffs	Domestic firms get protection.	Input costs may rise; retaliation risk exists.	Separate product tariffs from input tariffs.
Battery costs down	EV affordability improves.	Savings may be passed to consumers instead of shareholders.	Check who captures the cost reduction.

4. Robotics as the next stack

Robotics is best taught as a stack, not as a single product. An industrial robot, humanoid robot, warehouse robot, or autonomous vehicle requires mechanical systems, sensors, edge compute, AI models, training data, safety controls, battery/power systems, and customers with real tasks to automate.

The International Federation of Robotics reported that China represented 54% of global industrial robot deployments in 2024, with 295,000 installations and an operational robot stock above 2 million [S2]. This is important for students because China is not only a consumer market; it is a deployment environment, manufacturing base, and potential competitor.

Robotics layer	What it does	Companies or evidence to research
AI training	Teaches the model to perceive and decide.	Data-center GPUs, simulation, labeled data, software stack.
Edge compute	Runs inference on the device in real time.	Nvidia modules, custom chips, power/thermal constraints.
Sensors	Turns the physical world into data.	Cameras, radar, lidar, force sensors, IMUs.
Actuators	Turns decisions into motion.	Motors, gears, hands, grippers, torque control.
Battery/power	Determines runtime and economics.	Energy density, charging, maintenance cost.
Factory scale	Converts prototype into output.	Capex, yield, automation, supplier qualification.
Customer workflow	Defines whether the robot saves money.	Warehouses, factories, logistics, healthcare, construction.

5. Semiconductors: why the chip stack matters

In the conversation, GPUs, ASML, TSMC, and Nvidia came up because advanced robotics and autonomy are compute problems as much as mechanical problems. The lesson for students is that a robot company may depend on suppliers many layers away from the final product.

NVIDIA describes its accelerated computing platform as built around GPUs, CPUs, networking, software, and a physical AI platform for training, simulating, and deploying automation and robotics solutions [S5]. ASML is relevant because its lithography tools are essential to advanced chip manufacturing; ASML's 2025 annual report also pointed to strong EUV revenue expectations in 2026 [S6]. TSMC's investor page is a primary place for students to pull annual reports and Form 20-F filings when evaluating foundry exposure [S8].

Question	Why it matters	Evidence source
Who designs the chips?	Design leadership can create performance and software advantage.	Nvidia filings, Tesla filings, chip-company 10-Ks.
Who manufactures the chips?	Foundry bottlenecks can limit supply or raise costs.	TSMC annual report/Form 20-F.
Who makes the manufacturing equipment?	Advanced-node capacity depends on specialized equipment.	ASML annual report.
Where is packaging capacity?	AI accelerators need advanced packaging and memory integration.	Foundry and memory supplier filings.
What is restricted by policy?	Export controls and tariffs can change market access.	Federal Register, Commerce, company risk factors.

6. Industrial policy and tariffs

Industrial policy is not a side issue. It can determine where factories are built, which imports become expensive, which supply chains are subsidized, and which companies receive protection. In 2024, the Federal Register notice for Section 301 modifications listed tariff increases including electric vehicles to 100% in 2024, lithium-ion EV batteries to 25% in 2024, and semiconductors to 50% in 2025 [S3]. The U.S. Department of Commerce also describes CHIPS for America funding as aimed at domestic semiconductor manufacturing, research, and supply-chain resilience [S7].

Class discussion

Ask students: If a tariff protects a domestic company from foreign competition, does that automatically make the stock attractive? The answer is no. Protection can help pricing, but it can also raise input costs, slow adoption, or invite retaliation.

Policy tool	Possible benefit	Possible risk	What students check
Tariff on finished EVs	Protects domestic sellers from low-price imports.	Consumer prices may stay high; foreign firms may localize production.	Import exposure and domestic capacity.
Tariff on batteries	Encourages domestic battery supply.	Raises input cost for carmakers dependent on imports.	Supplier mix and pass-through ability.
CHIPS grants	Supports domestic fabs and R&D.	Project delays, cost overruns, political changes.	Award status, capex, construction progress.

Policy tool	Possible benefit	Possible risk	What students check
Export controls	Protects strategic technology.	Limits sales to certain regions; creates compliance risk.	Geographic revenue mix and risk factors.

7. Tesla case lab: from car story to manufacturing conversion

The transcript framed Tesla as more than a car company if it can convert manufacturing assets, engineering talent, AI infrastructure, and supply-chain relationships into robotics. The purpose of the case lab is not to tell students to buy or sell Tesla. It is to show them how to test a thesis.

Tesla's 2025 Form 10-K reported that R&D; expense increased 41% year over year, primarily due to costs related to AI and other programs as it expanded its product roadmap and technologies [S4]. The same filing reported capital expenditures for global AI and operational infrastructure and factory expansion [S4]. Those are evidence points students can use, but they are not enough by themselves. The thesis still requires proof of product readiness, manufacturing yield, margins, safety, regulation, and customer demand.

Claim	What would prove it?	What would weaken it?
Tesla can pivot from EVs to robotics.	Clear product specs, production ramp, customer pilots, margin path, safety record.	Repeated delays, unclear pricing, high support costs, no commercial demand.
Factories can be reused.	Shared manufacturing processes, supplier overlap, capex efficiency.	Robots require different tooling, labor, QA, or compliance.
AI infrastructure is a moat.	Lower training/inference cost, better real-world performance, proprietary data advantage.	Competitors access similar models, chips, or data.
Tariffs protect domestic robotics.	Foreign alternatives become costly and domestic demand remains strong.	Inputs get more expensive or production moves around tariffs.

8. Student workflow: build the stack before the thesis

Use this workflow any time a student says, 'This is the future.' The phrase may be true, but it is not yet an investment thesis.

Step	Student action	Output
1	Define the product clearly: EV, autonomous vehicle, humanoid robot, warehouse robot, chip, software platform.	One-sentence product definition.
2	Draw the value chain from raw inputs to end customer.	Layer map.
3	Find the bottleneck: supply, cost, regulation, safety, distribution, or customer ROI.	Bottleneck thesis.
4	Name the company that controls or benefits from the bottleneck.	Ticker shortlist.
5	Read primary filings and official data before opinions.	Evidence table.
6	Create bull, bear, and neutral scenarios.	Decision tree.
7	Decide whether the setup is trade, investment, or no action.	Action classification.

9. Workshop assignment

Assignment: Robotics value-chain memo

Write a two-page memo on one company in the robotics stack. The company does not have to be Tesla. It can be a chip designer, foundry, lithography company, sensor maker, industrial-automation company, battery supplier, or software platform.

Required sections

- Product and customer: What problem does the company solve?
- Value-chain position: Which layer does it control?
- Demand evidence: What official data supports the market?
- Margin evidence: How does the company make money?
- Policy exposure: What tariffs, subsidies, export controls, or safety rules matter?
- Bear case: What would prove the thesis wrong?
- Conclusion: trade, investment, watchlist, or avoid.

Scoring rubric

Category	Excellent	Needs work
Evidence	Uses filings, official reports, and direct company sources.	Relies mainly on YouTube, social posts, or price action.
Value chain	Clearly maps inputs, suppliers, customers, and policy.	Only describes the product.
Risk	Names specific margin, regulation, or supply-chain risks.	Says 'risky' without explaining why.
Decision	Explains why this is a trade, investment, watchlist, or avoid.	Jumps straight to buy/sell language.

10. Glossary

Term	Plain-English definition
Value chain	The chain of companies, inputs, factories, software, policy, and customers that make a product possible.
Industrial policy	Government action designed to shape strategic industries through funding, tariffs, rules, or procurement.
Tariff	A tax on imports that can protect domestic producers but may raise costs.
Foundry	A company that manufactures chips designed by other companies.
Lithography	The semiconductor manufacturing process that patterns tiny circuits onto wafers.
Physical AI	AI used in machines that act in the real world, such as robots, vehicles, and industrial automation systems.
Capex	Capital expenditures: money spent on factories, machinery, infrastructure, and long-term assets.

11. Source pack

ID	Source	How students should use it
T	April 21, 2026 conversation transcript	Conversation context: EVs versus robotics, Tesla manufacturing conversion, China/tariffs, and chip dependencies.
S1	IEA - Global Energy Review 2026: Electric vehicles	Use for global EV demand data and adoption framing.
S2	International Federation of Robotics - World Robotics 2025	Use for industrial robot deployment data and China context.
S3	Federal Register - Section 301 tariff modifications	Use for EV, battery, semiconductor, and critical-mineral tariff details.
S4	Tesla 2025 Form 10-K	Use for R&D, capex, AI, factory, and risk-factor evidence.
S5	NVIDIA 2026 Form 10-K	Use for accelerated computing and physical AI context.
S6	ASML 2025 Annual Report	Use for advanced semiconductor equipment context.
S7	U.S. Department of Commerce - Semiconductor Industry	Use for CHIPS for America and domestic supply-chain policy context.
S8	TSMC Annual Reports	Use for foundry-level primary-source research.
S9	NHTSA - Automated Vehicle Safety	Use when discussing SAE/NHTSA automation levels and safety framing.

Educational note: This tutorial is for research training only. It is not investment advice, and the company examples are case studies.